MASARYK UNIVERSITY

FACULTY OF INFORMATICS

Conflict-free colouring of planar graphs

Bachelor's Thesis

LUKÁŠ MÁLIK

Brno, Spring 2023

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Advisor: prof. RNDr. Petr Hliněný, Ph.D.

Department of Computer Science

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FACULTAS ARTIS INFORMATICÆ • UNIVERSITAS MASARYKIANA •

Declaration

Hereby I declare that this paper is my original authorial work, which I have worked out on my own. All sources, references, and literature used or excerpted during elaboration of this work are properly cited and listed in complete reference to the due source.

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Abstract

In this thesis, we will give an overview of a few specific variants of conflict-free coloring and how they relate to planar and outerplanar graphs. We will focus on the bounds of chromatic numbers for the discussed colorings on planar and outerplanar graphs, mainly demonstrating the currently best bounds along with an accompanying proof.

We will also show some open problems and present two original results in this thesis. One is proving an already established bound, but in an arguably simpler way using a less technical approach. The other one solves one of the open problems. More precisely we will improve a lower bound for the conflict-free coloring on closed neighborhood for planar graphs from three, which was formerly established, to four.

Keywords

Conflict-free coloring, proper conflict-free coloring, planar graphs, outerplanar graphs, neighborhoods, chromatic numbers

Contents

1 Introduction **1**

Introduction **1**
1.1 Outline 2
1.2 Preliminaries 3
1.2.1 Basic definitions of graph theory 3

2 Conflict-free coloring **6**
2.1 Chromatic numbers 8

3 Current bounds for CFC_c **11**
3.1 Upper bound for planar graphs 11
3.2 Upper bound for outerplanar graphs 11
3.3 Lower bounds for planar and outerplanar graphs . . . 12

4 New lower bound for CFC_c of planar graphs **15**

5 Current bounds for CFC_o and CFC_o* **18**
5.1 Lower bounds for planar and outerplanar graphs . . . 18
5.2 First good upper bound for planar graphs 19
5.3 Proof of upper bounds 6 and 5 for planar and outerplanar graphs 19
5.3.1 Discussion of possible improvements 23
5.4 Current best bounds 23

6 Bounds for $pCFC_c$ of planar and outerplanar graphs **24**
6.1 Upper bounds 24
6.2 Lower bounds 24

7 Bounds for $pCFC_o$ of planar and outerplanar graphs **25**
7.1 Bounds for outerplanar graphs 25
7.1.1 Lower bound 25
7.1.2 Upper bound 26
7.2 Lower bound for planar graphs 26
7.3 Upper bounds for planar graphs 28

8 Conclusion **30**

Bibliography **31**

List of Figures

3.1 Outerplanar graph G_1 with $\chi_{CFC_c}(G_1) = 3$. 12
3.2 Construction used by [3] to prove $\chi_{CFC_c}(G_1) \geq 3$ taken from the work [4]. 13

4.1 Graph X representing one of the constructions. 15
4.2 Graph X_3 composed of three copies of graph X from Figure 4.1. 16
4.3 Planar graph G_2 with $\chi_{CFC_c}(G_2) = 4$. Each edge is replaced by the gadget X_3 from Figure 4.2. 17

5.1 Planar and outerplanar graphs establishing currently highest known lower bounds. (inspired by [5, 3]) 18

7.1 Outerplanar graph C_5, that has $\chi_{pCFC_o} \geq 5$. 25
7.2 Planar graph G with $\chi_{pCFC_o} = 6$ (inspired by [3]). 26
7.3 Subgraph H of the graph G from Figure 7.2 (inspired by [3]). 27

1 Introduction

One of the most established and fundamental problems in graph theory is that of Coloring the vertices of a graph. There are many applications of this problem, and it forms a large part of contemporary graph theory. One of the most famous proofs in this area is that four colors always suffice to color a planar graph, which was an open problem for a long time.

Considering that this concept has been studied for more than a century, many variants of it have been researched. One such variant is the conflict-free coloring (CFC) of hypergraphs – coloring of the vertices such that in every hyperedge, there is at least one color appearing only on one vertex.

This variant of coloring was introduced in relation to the frequency assignment problem in the work of Even, Lotker, Ron, and Smorodinsky [1] in 2003. Ever since its appearance, it has attracted a fair amount of attention. Another application of CFC is for example: assigning frequency bands to cellular antennae, in battery consumption aspects of sensor networks, and in RFID protocols, which was studied by Shakhar Smorodinsky [2].

The definition of the conflict-free coloring of hypergraphs covers many more subvariants, especially the hyperedges of a hypergraph may represent neighborhoods of vertices in a graph, which will be exactly what this thesis will be dealing with.

More precisely, this thesis will study the standing lower and upper bounds of proper and improper conflict-free coloring on planar and outerplanar graphs on open (CFC_o) and closed (CFC_c) neighborhoods.

The concept of proper conflict-free coloring was first introduced by Igor Fabrici and other in [3], where they established all the currently best bounds.

The bounds for improper conflict-free coloring were studied in recent years in a few relevant articles, which also serve as the source and starting point for this thesis. The bounds were extensively researched in 2018 in a work of Zachary Abel, Victor Alvarez, Erik D. Demaine, Sándor P. Fekete, Aman Gour, Adam Hesterberg, Phillip Keldenich, and Christian Scheffer named Conflict-Free Coloring of Graphs [4].

There they introduced some important bounds from which some are still the best we have to this day (mainly the lower bounds). In their work, they have also managed to show some tight bounds. Bounds are tight if the lower bound equals the upper bound.

Their work was later improved upon by a few other articles that this thesis will discuss in depth later. Mainly the upper bounds were drastically improved by F. Huang, S. Guo, and J. Yuan in their article: Short Note on Open-Neighborhood Conflict-Free Colorings of Graphs [5]. Though the upper bounds established in that article give the best results so far, there are still new works coming out with, although with worse bounds, but arguably easier and more understandable proofs. See [6, 7, 3].

1.1 Outline

- Chapter 1 will continue with giving standard definitions, beginning with some basic concepts of graph theory and graph coloring.
- Chapter 2 will define conflict-free coloring along with the closed and open neighborhoods. It will give a brief overview of the past studies done on this coloring as well as some related variants of it, and give a comprehensive list of the best bounds.
- Chapters 3 and 5 will provide an overview of the relevant past results concerning the bounds on the improper conflict-free coloring of planar and outerplanar graphs with some brief mentions of the related proofs.
- Chapter 4 will feature our original work with a new result, along with the corresponding proof. Here we will show that the current lower bound for improper conflict-free coloring of planar graphs on the closed neighborhoods, which was until now 3, is not optimal, and we improve it to 4. (Theorem 4.0.3)
- Chapters 6 and 7 will provide an overview of the past results concerning the bounds on the proper conflict-free coloring of planar and outerplanar graphs with some brief mentions of the related proofs. It will also contain an original new proof

(Theorem 7.3.1) of the upper bound of 8 for the proper conflict-free coloring of planar graphs. This upper bound of 8 has already been proven [3], however, our proof is less technical, short and quite easy.

1.2 Preliminaries

Here we give the necessary definitions and tools:

1.2.1 Basic definitions of graph theory

Definition 1.2.1. (Simple graph): A simple graph $G = (V, E)$ is a pair of two sets, where V is a set of vertices called the vertex set and $E \subseteq \binom{V}{2}$ is a set of unordered pairs of vertices from V. We call such pairs edges. E is then called the edge set. Furthermore, we simplify and denote an edge u, v for vertices $u, v \in V$ as uv.

It is important to note that in this thesis all graphs will be simple graphs.

Definition 1.2.2. (Subgraph): Let H and G be simple graphs. H is a subgraph of G iff $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.

Definition 1.2.3. (Independent vertex set): An independent vertex set of a graph G is a set of vertices $V(G)$, such that no two vertices are adjacent in G.

Definition 1.2.4. (Walk): Let G be a graph. A walk is a sequence of edges $(e_1, e_2, ..., e_{n-1})$, such that there is a coresponding sequence of vertices $(v_1, v_2, ..., v_n)$, for which holds $e_i = (v_i, v_{i+1})$ for $i = 1, 2, ..., n - 1$.

Definition 1.2.5. (Trail): A trail is a walk, where all edges are unique.

Definition 1.2.6. (Path): A path is a trail, where all vertices are unique.

Definition 1.2.7. (Distance): The distance between two vertices v and u in a graph G is the number of edges in a shortest path connecting v and u.

We will denote the distance between two vertices v and u as $dist(v, u)$.

Definition 1.2.8. (Connected graph): A connected graph G is a graph, such that there exists a path from every vertex of G to any other vertex of G.

Definition 1.2.9. (2-connected graph): A 2-connected graph G is a graph, such that after removing any vertex of G the resulting graph G' is connected.

Definition 1.2.10. (Planar graph): A planar graph G is a graph, such that it can be embedded in the plane, which means it can be drawn in the plane without any of its edges crossing other edges or vertices.

Definition 1.2.11. (Plane graph): A plane graph is an embedding of a planar graph in the plane.

Definition 1.2.12. (Face): Assume a plane graph G. Then G divides the plane into connected regions called the faces of G.

Definition 1.2.13. (Outer face): An outer face is a face with unbounded area (face that extends to infinity).

Definition 1.2.14. (Outerplanar graph): An outerplanar graph G is a graph, such that it can be embedded into the plane in a way that each vertex is adjacent to the outer face.

Definition 1.2.15. (Bipartite graph): Bipartite graph G is such a graph, where its vertices can be divided into two disjoint sets of vertices U and S and every edge connects a vertex from one set to another (there are no edges connecting two vertices from the same set).

Definition 1.2.16. (Dual graph): The dual graph G of a planar graph H is a graph, such that it has a vertex for every face in H, and an edge between those vertices iff the two corresponding faces of H are adjacent.

Definition 1.2.17. (Coloring): Any assignment of labels (colors) to elements of a graph will be referred to as a coloring. For example, an assignment to vertices is called a vertex coloring and to edges is called an edge coloring.

In this thesis, we will consider only vertex colorings.

Definition 1.2.18. (Proper coloring): It is an assignment of colors to the vertices of the graph, such that no two adjacent vertices have the same color.

The adjective proper is often omitted in the literature.

Theorem 1.2.1 ([8, 9, 10])**.** *Every planar graph G is properly colorable by at most four colors.*

This theorem is also known as the 4-color theorem and later on, we will refer to it as such.

Theorem 1.2.2 (e.g. Observation in [11])**.** *Every outerplanar graph G is properly colorable by at most three colors.*

Theorem 1.2.3 ([12])**.** (*Kuratowski's Theorem*)*: A graph is planar if and only if it does not contain any subdivision of K_5 or $K_{3,3}$.*

Theorem 1.2.4 ([12])**.** *A graph is outerplanar if and only if it does not contain any subdivision of K_4 or $K_{2,3}$.*

Proof. Let G be an outerplanar graph. We create a new vertex v inside the outer face of G and connect all vertices of G to v creating G'. We can see that G' is a planar graph (G is an outerplanar graph so we can connect all its vertices to v without any crossings). If G contained K_4 or $K_{2,3}$, then G' would have to contain K_5 or $K_{3,3}$, however, that would mean G' is not a planar graph which is a contradiction. □

2 Conflict-free coloring

This chapter will formally define concepts such as improper conflict-free coloring, neighborhoods relating to these colorings, partial conflict-free coloring (CFC*), proper conflict-free coloring, unique color, and a chromatic number (χ) of conflict-free colorings. Next, it will give a brief overview of prior studies made on these topics and we will also mention a variant of coloring stemming from conflict-free coloring which is the unique-maximum coloring. Finally, there will be given a list of current best bounds, which will be a focal point of our work.

Definition 2.0.1. (Open neighborhood): The open neighborhood of a vertex v of a graph G is the subgraph of G induced by all vertices adjacent to v. (v is excluded in this case).

Definition 2.0.2. (Closed neighborhood): The closed neighborhood of a vertex v of a graph G is the subgraph of G induced by v and all vertices adjacent to v. (v is included).

Definition 2.0.3. (Unique color): A unique color with regard to the neighborhood is such a color that there it appears exactly one time in that said neighborhood.

We will denote the set of unique colors of a vertex a as $\mu(a)$.

Definition 2.0.4. (Improper conflict-free coloring): An improper conflict-free coloring of a graph G with respect to open/closed neighborhood, or for short CFC_o/CFC_c, is a coloring of vertices (there can be two adjacent vertices sharing a color), such that for every vertex V of G, there is a unique color in its open/closed neighborhood.

Definition 2.0.5. (Proper conflict-free coloring): A proper conflict-free coloring of a graph G with respect to open/closed neighborhood, or for short $pCFC_o/pCFC_c$, is a proper coloring of vertices (there cannot be two adjacent vertices sharing a color), such that for every vertex V of G, there is a unique color in its open/closed neighborhood.

Note that when we will be speaking about conflict-free coloring and using the abbreviation CFC we will be referring to the improper one.

This thesis will denote which kind of neighborhood is used for the specific CFC as an index of the said CFC. More precisely we define the CFC_c as conflict-free coloring on closed neighborhoods and CFC_o as conflict-free coloring on open neighborhood.

Definition 2.0.6. (Partial conflict-free coloring): A partial conflict-free coloring is such a CFC that not every vertex V of graph G has to be colored - it can be left with unassigned color, but every vertex, including the uncolored ones, has to have a unique color in its close/open neighborhood.

We will denote this type of coloring with $*$, for example, CFC_c*. It is also used in some literature that the uncolored vertices bear the color zero.

Observation 2.0.1. *If we consider a graph G to have $CFC*$ α on an arbitrary neighborhood with x colors. Then we can say that G has a CFC δ on the same kind of neighborhood with $x + 1$ colors.*

To get δ we simply take the coloring α of G and color the uncolored vertices in G by color $x + 1$. Since we used a color that was not previously present in coloring α of G all the vertices retain the previous unique color they had in α. Therefore we can see that δ is indeed a correct CFC of G.

Definition 2.0.7. (Conflict-free chromatic number): Conflict-free chromatic number is the minimum number k such that a graph G admits a given kind of CFC using k colors.

The notation for chromatic number will be used in a way such that the coloring to which the chromatic number refers to, will be written as its index. For example χ_{pCFC_o} refers to a chromatic number of proper conflict-free coloring on an open neighborhood.

These colorings have been studied extensively already and there are many results concerning the NP-hardness and completeness of the colorings named above. Nice results showing that determining whether a graph admits a certain CFC_c is NP-complete is studied in the work of Luisa Gargano and Adele A. Rescigno [13], it was also examined in [4] they also dive into the problematic of completeness studying it for planar, outerplanar and bipartite graphs along with

some special cases. This thesis, however, will not cover the complexity of these problems.

There are also other related kinds of colorings, but one that provides more structure and is often used to prove other results in CFC setting is unique maximum coloring (UMC).

Definition 2.0.8. (Unique maximum coloring): A unique maximum coloring of a graph G with respect to an open/closed neighborhood, or for short UMC_o/UMC_c is the coloring of vertices such that for every vertex v of G its maximum color (we color by numbers, so the maximum color is the largest number) in its open/closed neighborhood is its unique color (this is a definition for improper setting).

Note that UMC also has both improper and proper variants (later denoted as UMC for improper and $pUMC$ for proper).

Observation 2.0.2. *For every planar graph G we have $\chi_{pCFC_o}(G) \leq \chi_{pUMC_o}(G)$.*

This type of coloring was already mentioned in 1995 as order coloring [14] and later it was studied by many others. Specifically, the relationship between CFC and UMC was examined by Panagiotis Cheilaris and Géza Tóth in [15]. What could be maybe more interesting in relation to this thesis, are the research results in [3] where they looked for the best bounds for both improper and proper χ_{UMC} on both neighborhoods on both outerplanar and planar graphs. Later we will not be frequently mentioning many findings relating to this topic as I failed to find any big connections or improvements to the bounds, but they are very similar, and further study may lead to surprising improvements.

2.1 Chromatic numbers

This thesis will be primarily focused on studying the lower and upper bounds of the CFC and $pCFC$ planar and outerplanar graphs. Currently, best results for these bounds are as follows:

Closed neighborhood:

1. For planar graphs: $3 \leq \chi_{CFC_c}(P) \leq 4$.

2. For outerplanar graphs: $\chi_{CFC_c}(O) = 3$.

Open neighborhood:

3. For planar graphs: $4 \leq \chi_{CFC_o}(P) \leq 5$.

4. For outerplanar graphs: $3 \leq \chi_{CFC_o}(O) \leq 4$.

In [3] it was conjectured that the lower bound for $\chi_{CFC_C}(P)$ (see point 1) which is 3 is the best and it will be necessary to improve the upper bound of 4. However, later in Chapter 4, we will show that the lower bound is actually 4 and thereby making the bound tight $\chi_{CFC_C}(P) = 4$.

There were many that have contributed to the research of these bounds. Namely in the work of [4] the authors discussed among many other things the bounds for CFC_c and they also have achieved some results regarding bipartite graphs. Later there was a nice result presented in [5] where the authors establish both the lower and upper bounds for CFC_o of both planar and outerplanar graphs. They actually established stronger results than the one given above more precisely they showed that the bounds of CFC_o* are tight. After, this discovery as mentioned in the introduction followed [6] where they achieved weaker results showing the upper bound of CFC_o for both planar and outerplanar graphs are 6 and 5. However, their proof is very short and comprehensive and will be presented in Chapter 5 in greater detail as I believe there are many good ideas hidden in it and room for improvement or at least for inspiration.

The bounds given by [5] on CFC_o* are:

5. For planar graphs: $\chi_{CFC_o*}(P) = 4$.

6. For outerplanar graphs: $\chi_{CFC_o*}(O) = 3$.

Then by Observation 2.0.1, we get the current bounds for CFC mentioned above.

Next, we present the bounds concerning $pCFC$ as follows:

Closed neighborhood:

7. For planar graphs: $\chi_{pCFC_o}(P) = 4$.

8. For outerplanar graphs: $\chi_{CFC_o}(O) = 3$.

Open neighborhood:

9. For planar graphs: $6 \leq \chi_{pCFC_o}(P) \leq 8$.
10. For outerplanar graphs: $\chi_{pCFC_o}(P) = 5$.

All the bounds for $pCFC$ were given in the work of Igor Fabrici, Borut Lužar, Simona Rindošová, Roman Soták [3] as the concept of $pCFC$ itself was also introduced by them in this paper. To prove these bounds they used a concept defined as facial closure some partition and the 4-color theorem. They also managed to prove many of the bounds regarding the UMC and $pUMC$. Later, more studies have been made examining $pCFC$ (see [16]), but none of them managed to improve the bound. However, the work done by Eun-Kyung Cho and others in [17] was able to show that if the graph fulfills some conditions the bound can be improved.

Bounds for UMC and $pUMC$ for closed neighborhood:

11. For planar graphs: $4 \leq \chi_{UMC_c}(P) \leq 6$.
12. For outerplanar graphs: $3 \leq \chi_{UMC_c}(O) \leq 5$.
13. For proper UMC on planar graphs: $6 \leq \chi_{pUMC_c}(P) \leq 8$.
14. For proper UMC on outerplanar graphs: $\chi_{pUMC_c}(O) = 5$.

Open neighborhood:

15. For planar graphs: $5 \leq \chi_{UMC_o}(P) \leq 10$.
16. For outerplanar graphs: $4 \leq \chi_{UMC_o}(O) \leq 5$.
17. For proper UMC on planar graphs: $6 \leq \chi_{pUMC_o}(P) \leq 10$.
18. For proper UMC on outerplanar graphs: $\chi_{pUMC_o}(O) = 5$.

All these bounds except one were provided in [3]. The lower bound $3 \leq \chi_{UMC_c}(O)$ was shown in [4].

It is also interesting to point out that these bounds still leave large room for improvement and can be the focus of future studies mainly the bound for $pUMC_o$ currently stands as $6 \leq \chi_{pUMC_o}(P) \leq 10$ and it was conjectured by [3] that this single lower bound based on their experiments is the only improvable one.

3 Current bounds for CFC_c

This chapter will give full proofs of the upper and lower bounds of CFC_c for planar and outerplanar graphs. Here we will use the original example for the lower bound of 3 for both outerplanar and planar graphs which is the best so far, and use similar argumentation as was implied in [4] for upper bounds.

3.1 Upper bound for planar graphs

Theorem 3.1.1 (Inspired by [4]). *For every planar graph G we have: $\chi_{CFC_c}(G) \leq 4$.*

Proof. By applying the 4-color Theorem 1.2.1 we get a coloring of G, such that there are no two vertices of the same color next to each other. Next, since we are speaking about the closed neighborhood, we just assume that for every vertex its unique color is the color the vertex itself holds. □

Note that this proof also holds for proper conflict-free coloring of planar graphs as we use the 4-color theorem, which uses classical proper coloring, so it is impossible for G to have two vertices u and v such that if they share an edge ($uv \in E(G)$) they share the same color.

3.2 Upper bound for outerplanar graphs

Theorem 3.2.1 (Inspired by [4]). *For every outerplanar graph G we have: $\chi_{CFC_c}(G) \leq 3$.*

Proof. By applying the Theorem 1.2.2 we get a coloring of G, such that there are no two vertices of the same color next to each other. Next, since we are talking about the closed neighborhood, we just assume that for every vertex its unique color is the color the vertex itself holds. □

Note that again this proof can be extended to the proper conflict-free coloring of outerplanar graphs using the same arguments as before.

3.3 Lower bounds for planar and outerplanar graphs

Theorem 3.3.1 (Inspired by [4]). *There exists an outerplanar graph G, such that: $\chi_{CFC_c}(G) \geq 3$.*

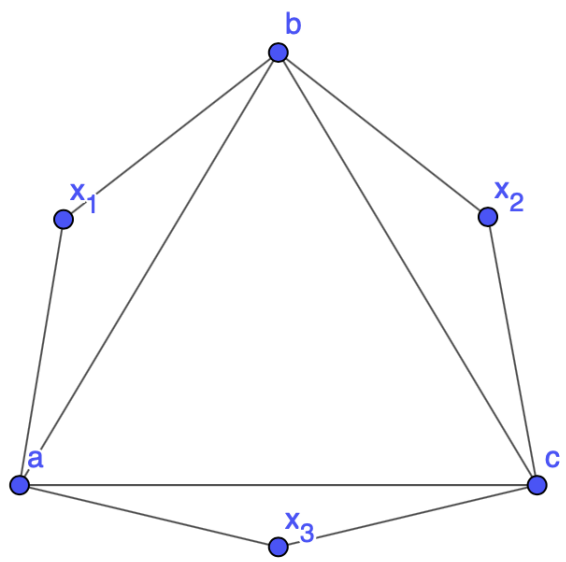


Figure 3.1: Outerplanar graph G_1 with $\chi_{CFC_c}(G_1) = 3$.

Proof. Let G_1 be the graph from Figure 3.1. We suppose to the contrary that there is a CFC_c of graph G_1 such that it uses only 2 colors. Now we will consider the cases regarding the coloring of vertex a. We will also denote α as CFC_c of a graph G_1.

Since vertex a has to have a color we can without any loss of generality assume that $\alpha(a) = 1$. a has two options where its unique color can be. either it is the $\alpha(a)$ itself or a coloring of some neighboring vertex (by another color than $\alpha(a)$ since we are considering a close neighborhood).

Case 1: If $1 \in \mu(a)$ then all its neighbors would have to be colored by 2. If we do so however then the remaining uncolored vertex x_2 cannot be colored by 1, because then a vertex c, as well as b, would not have a unique color and therefore it would not be a valid CFC_c and

$\alpha(x_2) = 2$ is also invalid as in this case, x_2 would not have a unique color as all its neighbors are colored by 2.

Case 2: If $2 \in \mu(a)$, then only one of its neighbors can be colored by 2. without a loss of generality we assume that the color 2 is on vertex x_3 or x. b and x_2 therefore have to be colored by 1 and we can already see that x_2 has no unique color in this case. Thus by symmetry, we know that this case is invalid.

Hence we get that in both cases it does not admit to CFC_c by 2 colors, and since every outerplanar is also a planar graph we get the following lower bound for both instances: $\chi_{CFC_c}(G_1) \geq 3$. □

In the study done in [3] the authors have looked into more complex graphs in hopes of finding better lower bounds for planar graphs taking inspiration from some graphs, shown in [4] and trying to experiment with them. They showed the following construction in their articles:

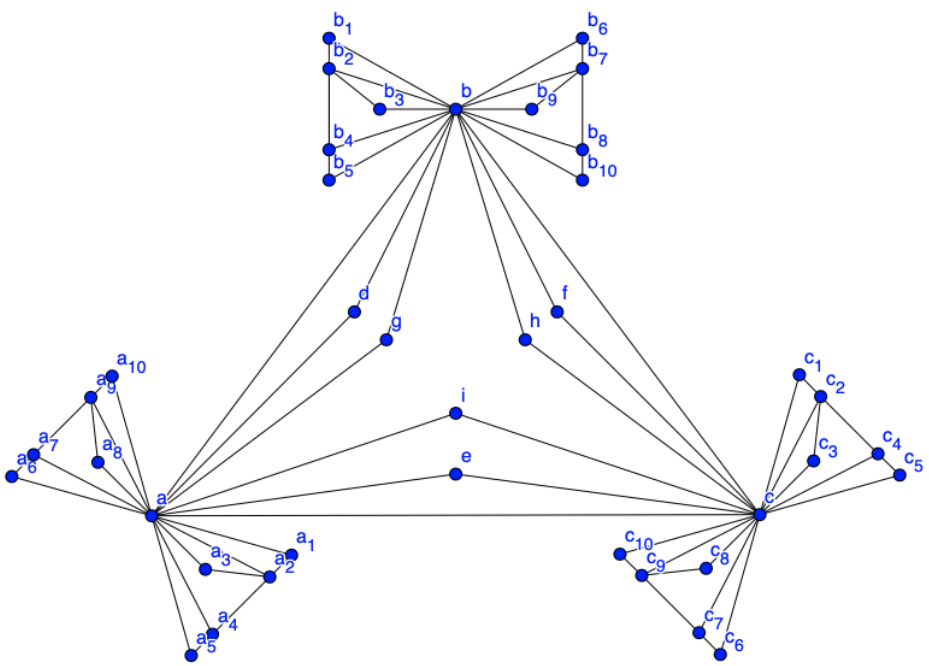


Figure 3.2: Construction used by [3] to prove $\chi_{CFC_c}(G_1) \geq 3$ taken from the work [4].

However, this graph admits CFC_c α of 3 colors as they have found out. We just color $\alpha(a) = \alpha(b) = \alpha(c) = 2, \alpha(a_{10}) = \alpha(b_6) = \alpha(c_1) = 3$ and all other by 1.

Finally, they conjectured, as mentioned before, that the lower bound is correct. In the next chapter however, we will give an example of a graph that is not CFC_c with 3 colors and therefore prove that the conjectured lower bound of 3 was not optimal and is improvable to 4.

4 New lower bound for CFC_c of planar graphs

This chapter will show the original result of this thesis, proving that there exists a planar graph G such that $\chi_{CFC_c}(G) \geq 4$. Recall that the previous best was 3 presented in [4] and [3].

The proof will be done by means of contradiction. First, we will look at some simple constructions and prove a few lemmas about their special properties and after that, we will use these gadgets to build the final graph that will not be conflict-free colorable by 3 colors.

Lemma 4.0.1. *Consider the graph X of Figure 4.1. If vertices a and b share the same color, then the X does not admit CFC_c using 2 colors.*

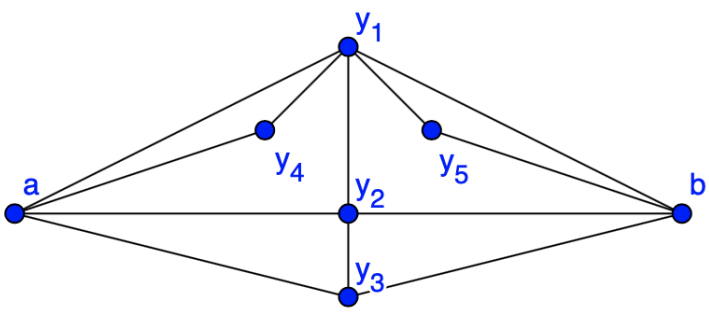


Figure 4.1: Graph X representing one of the constructions.

Proof. Let X be a graph from Figure 4.1. We suppose to the contrary that X admits CFC_c α using at most 2 colors while a and b share the same color. Without any loss of generality, we assume $\alpha(a) = \alpha(b) = 1$.

Here we will consider two cases. First one when y_1 is colored by 1, and second when $\alpha(y_1) = 2$.

Case 1: In this instance, it is impossible to color vertices y_4 and y_5 in such a way that either they or y_1 have a unique color. Neither of them can be colored by 1, because then they would not have a unique color as all its neighbors namely a, b, y_1 have the same color 1. Furthermore, they also cannot admit to $\alpha(y_4) = \alpha(y_5) = 2$ as then the vertex y_1 would not have a unique color. Hence we get that y_1 cannot be colored by 1 (or, in other words, the same color as a and b).

Case 2: Here we consider the coloring $\alpha(y_1) = 2$. For y_1 to have a unique color it has to be color 2 (because a and b are already colored by 1), which is the color of the vertex itself. From this, we get that all its other neighbors must be colored $\alpha(y_2) = \alpha(y_4) = \alpha(y_5) = 1$. Finally, we have only y_3 which is left uncolored, but this vertex cannot be colored by 1 as all its neighbors are colored by 1 and so it would not have a unique color and it also cannot be colored by 2 as then the vertex y_2 would not have a unique color namely vertices y_1, y_3 would be colored by 2 and vertices a, b, y_2 would be colored by 1.

So we get that y_1 can be colored neither by 1 nor 2 which is a contradiction. □

Observation 4.0.1. *When considering the construction X and its CFC_c α, where vertices a, b share the same color, then a or b or both together are neighbors of at least two colors different from $\alpha(a) = \alpha(b)$. (observation from the proof of the previous lemma 4.0.1)*

Lemma 4.0.2. *Consider the graph X_3 of Figure 4.2 and the CFC_c of X_3 using 3 colors, then a and b cannot share the same color.*

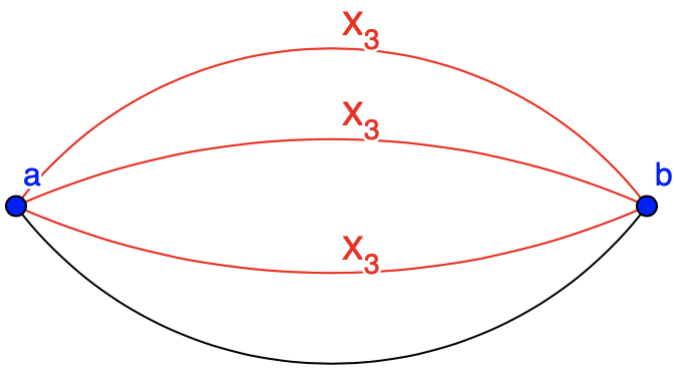


Figure 4.2: Graph X_3 composed of three copies of graph X from Figure 4.1.

Proof. Let X_3 be a graph from Figure 4.2. We suppose to the contrary that a and b can share the same color and X_3 still admits to the CFC_c using 3 colors. Without any loss of generality, we will assume that $\alpha(a) = \alpha(b) = 1$.

Next, from Observation 4.0.1 we get that a or b or both together have at least two neighbors of color 2 and two neighbors of color 3 (combination of Observation 4.0.1 and the fact that we are using only 3 colors). However, since a and b also share an edge then we are getting that at least one of them cannot have a unique color, hence α is not a valid CFC_c, thus getting our contradiction. □

Theorem 4.0.3. *For every planar graph G we have: $\chi_{CFC_c}(G) \geq 4$.*

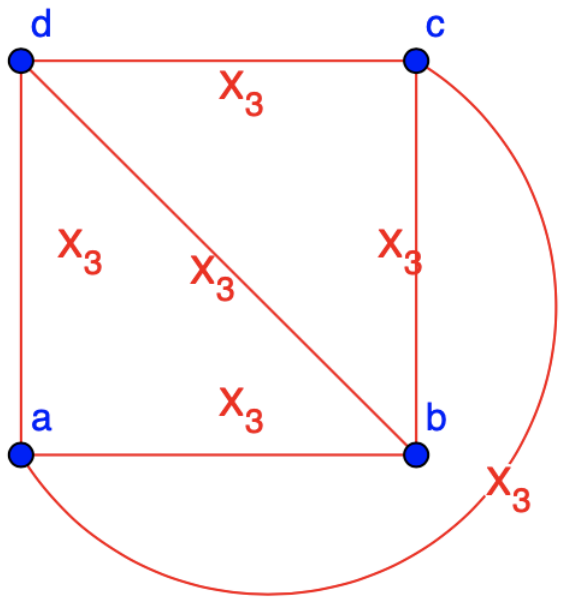


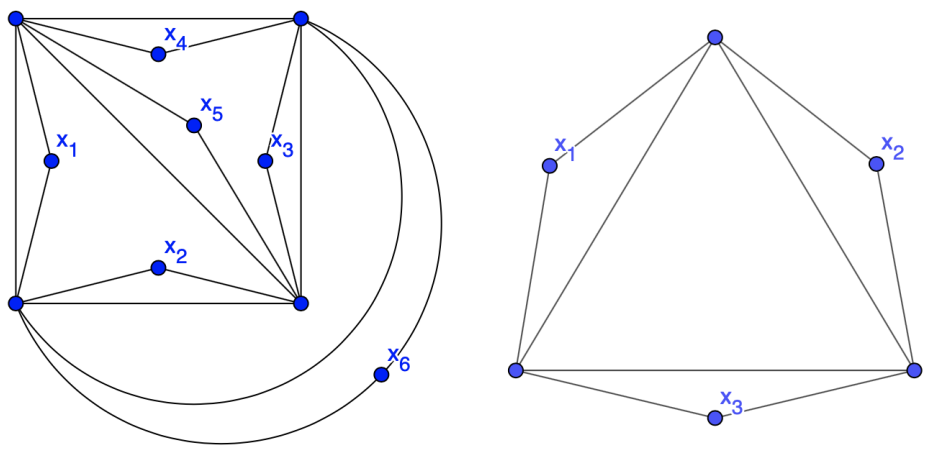
Figure 4.3: Planar graph G_2 with $\chi_{CFC_c}(G_2) = 4$. Each edge is replaced by the gadget X_3 from Figure 4.2.

Proof. Let G_2 be a graph from Figure 4.3. Suppose to the contrary that it admits a CFC_c by 3 colors. Since the underlying structure of G_2 is a K_4 with all edges replaced by X_3, and we are assuming CFC_c with 3 colors, then from lemma 4.0.2 we know that this is equivalent to the problem of the classical proper coloring of K_4 with 3 colors and we know that this is impossible since K_4 has no proper 3-coloring. □

5 Current bounds for CFC_o and CFC_o*

This chapter will show examples of graphs giving lower bounds for CFC_o. We will also look into some interesting proofs concerning the upper bounds, especially the proof showing that $\chi_{CFC_o}(G) \leq 6$ by [6]. Even though this bound is not the best one, we find this proof to be very comprehensive with room for potential improvement. There will be also given some ideas that we tried with hopes of lowering the bound using this proof as a starting point.

5.1 Lower bounds for planar and outerplanar graphs



(**a**) Planar graph G_1, which has $\chi_{CFC_o} = 4$.

(**b**) Outerplanar graph G_2, which has $\chi_{CFC_o} = 3$.

Figure 5.1: Planar and outerplanar graphs establishing currently highest known lower bounds. (inspired by [5, 3])

Observation 5.1.1. *When considering CFC_o, vertices of degree 2 must be adjacent to two different colors, otherwise, they do not see a unique color.*

From previous Observation 5.1.1 we can see that vertices x_y for $y = \{1, 2, 3, 4, 5, 6\}$ in a planar graph G_1 from Figure 5.1a and x_y for $y = \{1, 2, 3\}$ in an outerplanar graph G_2 from Figure 5.1b make the

problem of CFC_o these graphs equivalent to the proper coloring of K_4 and K_3, which as we know require 4 and 3 colors respectively.

5.2 First good upper bound for planar graphs

One of the earliest bounds was given in work [4] which established that 8 colors always suffice on planar graphs.

Definition 5.2.1. (Independent domination set): Let G be a graph, then the independent domination set is an independent vertex set $S \subseteq V(G)$, such that every vertex $v \in V(G)$ either is in S or is adjacent to a vertex in S.

Theorem 5.2.1 ([4])**.** *For every planar graph G we have $\chi_{CFC_o} \leq 8$.*

Proof. Let G be a planar graph. Firstly we construct an independent domination set U of G. Let $U_1 = V(G) \backslash U$. Next, we construct a minor G_1 of G by contracting every vertex of V into its arbitrary neighbor belonging to V_1. Similarly, we construct the minor G_2 of G by contracting every vertex of V_1 into its arbitrary neighbor belonging to V. Since both G_1 and G_2 are planar we then apply a 4-color Theorem 1.2.1 to both of them coloring them by $\{1, 2, 3, 4\}$ and $\{5, 6, 7, 8\}$ respectively.

Next, we extend this coloring to G assigning every vertex of G color it has in either G_1 or G_2 (G_1 and G_2 are disjoint). This coloring is indeed a valid CFC_o and the unique color of every vertex is the color of the vertex it was contracted into (by contracting a vertex v into its n neighbor u we make all neighbors of v adjacent to u. Then by using proper coloring, and extending it to the original, the color assigned to u is unique in the neighborhood of v). □

5.3 Proof of upper bounds 6 and 5 for planar and outerplanar graphs

This proof was presented in [6]. Their solution is very elegant showing a relatively strong result. The authors proved that the upper bound for CFC_o* is 5 for planar graphs and 4 for outerplanar graphs. Then by applying the Observation 2.0.1 we get that the upper bounds for CFC_o are 6 for planar graphs and 5 for outerplanar graphs.

Their proof first starts with defining a special set called a 3-distance set, then the authors speak about how to construct this set in a given planar graph G which they then partition into 3 special sets of vertices. Then, we construct graph G' from G followed by some contractions between distinct sets. Finally, they color these sets of vertices with 5 distinct colors leaving some uncolored and proof that the said assignment of colors is indeed a valid CFC_o* coloring. Next, we will go through each of these steps:

Definition 5.3.1 ([6])**.** (maximal 3-distance set S): For a graph G a maximal 3-distance set S is a set of vertices $S \subseteq V(G)$, such that it satisfies the following:

- Distance between every pair of vertices $v, v' \in S$ is at least 3. $dist(v, v') \geq 3$.
- $\forall v \in S, \exists v' \in S$, such that $dist(v, v') = 3$.
- $\forall u \notin S, \exists u' \in S$, such that $dist(u, u') < 3$.

Let G be a connected planar graph, to construct S in G we first pick an arbitrary vertex v in G. We then iteratively and greedily add a vertex $u \in V(G)$ to S if two conditions are fulfilled:

1. $\forall w \in S$, it holds that $dist(w, u) \geq 3$.
2. $\exists w \in S$, such that $dist(w, u) = 3$.

We continue to repeat this iterative process until no more vertices can be added.

By doing the above-described steps we have managed to construct a maximal 3-distance set S in G. Note that the S is an independent set by definition since all its vertices are not connected. We then build two other sets namely S_1 and S_2. Set S_1 will represent all the neighbors of the vertices present in S. Formally: $S_1 = \{v|(v, u) \in E(G), u \in S\}$. Set S_2 will denote all other vertices not present in neither S or S_1. $S_2 = V(G)\ (S \cup S_1)$.

Now we will summarize what we know about the 3 sets we constructed from connected G in a few observations.

Observation 5.3.1. *S is an independent set.*

Observation 5.3.2. *$\forall u \in S_1$ there exists exactly 1 vertex $v \in S$, such that $(u, v) \in E(G)$.*

This is because if there were two vertices $v \in S$ and $v' \in S$ such that $(u, v) \in E(G)$ and $(u, v') \in E(G)$. Then $dist(v, v') = 2$, which we know cannot happen thanks to how we constructed the set S. There also has to be at least one vertex $v \in S$ such that $(u, v) \in E(G)$, because of how we define the set S_1.

Observation 5.3.3. *$\forall v \in S$ there exists at least one vertex $u \in S_1$, such that $(v, u) \in E(G)$. Otherwise, G consists of a single vertex.*

If there was a vertex $v \in S$, such that it has no neighbor $u \in S_1$, then v is clearly an isolated vertex, and since we assumed G to be a connected planar graph, this case can happen only if G has a single vertex.

Observation 5.3.4. *$\forall w \in S_2$ there exist no vertex $v \in S$, such that $(w, v) \in E(G)$.*

This is because if there was such an edge then vertex w would belong to a vertex set S_1.

Observation 5.3.5. *$\forall w \in S_2$ there exists at least one vertex $u \in S_1$, such that $(v, u) \in E(G)$.*

From this, we can also get the following observation:

Observation 5.3.6. *$\forall w \in S_2$ there exists a vertex $v \in S$, such that $dist(w, v) = 2$.*

Next, we will construct a graph G'. For this, we will first remove all edges between any two vertices of S_2. From this, we get that both S and S_2 have only neighbors from the vertex set S_1. By Observation 5.3.3 and Observation 5.3.5 we also know that all have at least one neighbor in S_1. We then contract every vertex $x \in (S \cup S_2)$ into an arbitrary vertex $u \in S_1$, such that $(x, u) \in E(G)$. This gives us a graph G' for which it holds that $V(G') = S_1$.

Finally, we can prove the following theorems:

Theorem 5.3.1 ([6]). *For every planar graph G we have $\chi_{CFC_o*} \leq 5$.*

Proof. Let G be a connected planar graph. Firstly we construct the 3-distance set S and divide the vertices into 3 sets S, S_1 and S_2 as described above. Following that we build the graph G' based on the instructions given above. Since we only used edge contraction and deletion, G' stays planar. We then apply the 4-color Theorem 1.2.1 to G', giving us a coloring α using four colors $\{1, 2, 3, 4\}$.

Next, we extend this coloring α to the graph G resulting in all vertices in set S_1 of G colored. Finally, we color vertices of set S of G by 5 and leave vertices in S_2 uncolored. We argue that the coloring we described here is a valid CFC_o* using 5 colors.

If we consider vertex $x \in (S \cup S_2)$, its unique color would be on the vertex that we contracted it into when constructing the graph G'. This is because when we contract the vertex, all its other neighbors will then also become the neighbors of the vertex we contracted it into (when considering contracting by an edge), and since G' was then colored using proper coloring, then no two neighbors share a color.

For every vertex $u \in S_1$ it holds that its unique color is assigned to the vertex of $v \in S$, which they are adjacent to. This is because the Observation 5.3.2 tells us that every vertex u has exactly one neighbor from set S and thanks to the fact they have been assigned color 5.

Since we have shown that all sets of vertices have a unique color and we get that this coloring is indeed a valid CFC_o* of G. Finally let H be a planar graph by applying the above-described coloring to each of its connected components we get the valid CFC_o* of H proving the theorem. □

Theorem 5.3.2 ([6]). *For every outerplanar graph G we have $\chi_{CFC_o*} \leq 4$.*

Proof. Let G be an outerplanar graph. We can do the exact same steps as we did before with the only difference being, that when we construct the G' we can use the Theorem 1.2.2 using only 3 colors $\{1, 2, 3\}$, giving us the coloring α. We can do this because again to construct G' we use only edge deletion and contraction, which are operations that preserve outerplanarity.

Then, we again extend this coloring to G and use the fourth color for vertices in set S. By applying the same arguments as before we can see that this is indeed a valid CFC_o* of G. □

By Observation 2.0.1 we get the following bounds [6]:

- For planar graphs $\chi_{CFC_o} \leq 6$.
- For outerplanar graphs $\chi_{CFC_o} \leq 5$.

5.3.1 Discussion of possible improvements

We find the construction in this proof very elegant and tried quite a few variants to improve the results. One of the tools we have tried to use repeatedly is as follows:

Conjecture 5.3.3. *Let G be a connected planar graph. If we remove the edges between all vertices of S_2 and contract them to their arbitrary neighbor in S_1, then by removing vertices in S we get a graph G' that is composed of connected components that are outerplanar graphs.*

This conjecture turned out to be wrong, however, as was later found out. We can actually construct a graph and a valid partition of vertices, such that after removing the vertices in S and contraction of vertices in S_2, the remaining graph G' would not be outerplanar.

One of the ideas we tried is also to incorporate a concept known as facial closure into the proof. This concept was mentioned and explained in [3]. Instead of contracting the vertices of S and S_2 into S_1 we would do a facial closure on these sets and study the resulting graph. This resulted in interesting graphs. However, even in this way, we did not manage to improve the bounds or shorten the proof.

5.4 Current best bounds

Finally, the best current upper bounds are given in [5] which proves the following Theorems:

Theorem 5.4.1 ([5])**.** *For every planar graph G we have $\chi_{CFC_o}* \leq 4$.*

Theorem 5.4.2 ([5])**.** *For every outerplanar graph G we have $\chi_{CFC_o}* \leq 3$.*

These results however, will not be presented in this work as the proof is very technical and relies heavily on matching theory [18], therefore is beyond the scope of this thesis.

6 Bounds for $pCFC_c$ of planar and outerplanar graphs

This chapter will quickly summarize what we know about the bounds for $pCFC_c$ (the proper-coloring version).

6.1 Upper bounds

Theorem 6.1.1 ([3]). *For every planar graph G we have $\chi_{pCFC_c} \leq 4$.*

Proof. Let G be a planar graph. We properly color G by 4 colors using 4-color Theorem 1.2.1. This coloring is also a valid $pCFC_c$ since every vertex has a unique color which is the color assigned to the vertex itself. □

Theorem 6.1.2 ([3]). *For every outerplanar graph G we have $\chi_{pCFC_c} \leq 3$.*

Proof. Similarly, we let G be an outerplanar graph. We properly color G by 3 colors using Theorem 1.2.2. This coloring is also a valid $pCFC_c$ since every vertex has a unique color which is the color assigned to the vertex itself. □

6.2 Lower bounds

The lower bound for a planar graph is given by K_4, which since we are considering $pCFC_c$, that is a proper coloring, needs exactly 4 colors to be fully colored.

The lower bound for an outerplanar graph is given by K_3, which again needs exactly 3 colors to be fully colored.

We can see that by combining the findings about lower and upper bounds gives us the tight bounds [3] of $\chi_{pCFC_c} = 4$ for planar graphs and $\chi_{pCFC_c} = 3$ for outerplanar graphs.

7 Bounds for $pCFC_o$ of planar and outerplanar graphs

This chapter will firstly show the tight bounds of outerplanar graphs for $pCFC_o$ as shown in the work of Igor Fabrici and others [3], then we will look also into their proof of lower bound of planar graphs for $pCFC_o$ shown in the same study. Lastly, we will only briefly summarize their idea about the upper bound of planar graphs for $pCFC_o$ and then show the original new proof of the upper bound, which is done in a simpler way with a less technical approach, but still gives the same result of $\chi_{pCFC_o} \leq 8$, which is the current best bound.

7.1 Bounds for outerplanar graphs

7.1.1 Lower bound

Theorem 7.1.1 ([3]). *For every outerplanar graph G we have $\chi_{pCFC_o} \geq 5$.*

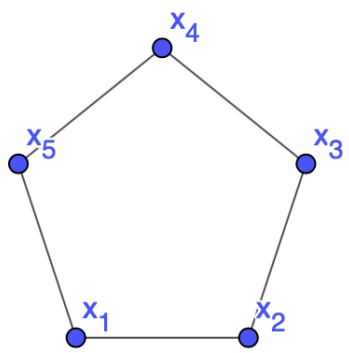


Figure 7.1: Outerplanar graph C_5, that has $\chi_{pCFC_o} \geq 5$.

Proof. Let C_5 be the graph from Figure 7.1. We assume to the contrary that C_5 is colorable by 4 colors.

We can see that there has to be at least one color that appears twice in C_5. Since we are considering a proper conflict-free coloring we do not allow two same colors to be present on vertices next to each other.

In C_5 there cannot be vertices apart from each other at distance higher than two. From this, we can consider that the two vertices colored by the same color are apart from each other at distance two. However, if they are at distance two from each other then the vertex between them does not have a unique color and therefore it is not a valid $pCFC_o$, hence we get that C_5 is not colorable by 4 or less colors, which is our contradiction. □

Note that we can color C_5 by 5 colors by simply assigning each vertex a different color.

7.1.2 Upper bound

Theorem 7.1.2 ([3]). *For every outerplanar graph G we have $\chi_{pCFC_o} \leq 5$.*

In work [3] the authors show the upper bound for $pUMC_o$ for outerplanar graphs, more precisely they showed that $\chi_{pUMC_o} \leq 5$. Then by Observation 2.0.2 they got the the bound for $pCFC_c$ of $\chi_{pCFC_c} \leq 5$. Their proof for $pUMC_o$ is however quite technical and will not be shown in this thesis.

7.2 Lower bound for planar graphs

This example was shown in [3].

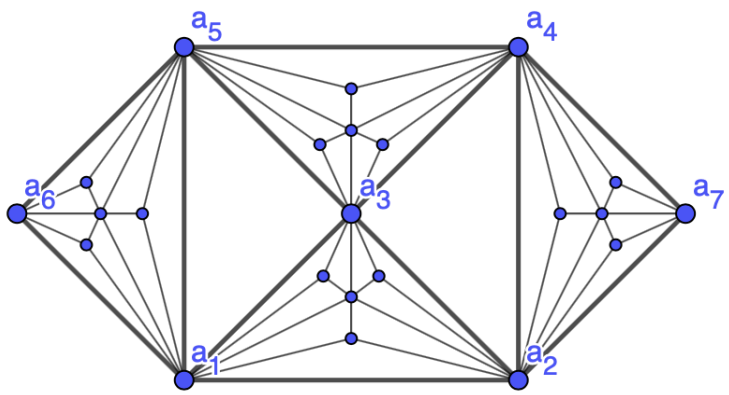


Figure 7.2: Planar graph G with $\chi_{pCFC_o} = 6$ (inspired by [3]).

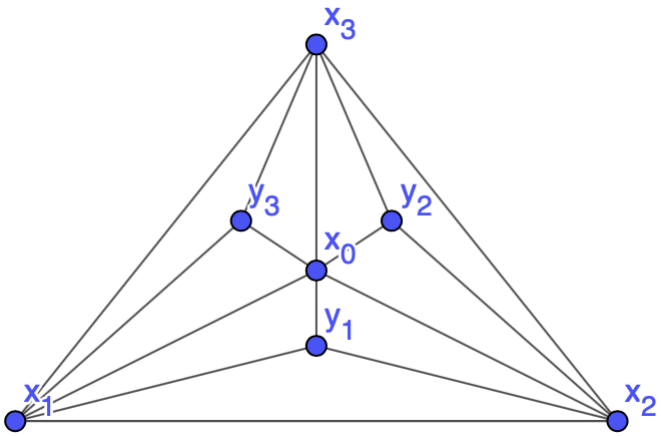


Figure 7.3: Subgraph H of the graph G from Figure 7.2 (inspired by [3]).

Theorem 7.2.1 ([3]). *There exists a planar graph G, such that $\chi_{pCFC_o}(G) \geq 6$.*

Proof. Let G be the planar graph from Figure 7.2. We suppose to the contrary that G admits a $pCFC_o$ α with at most 5 colors. Consider the Subgraph H of G depicted in Figure 7.3. Since we are considering proper coloring and the vertices x_0, x_1, x_2 and x_3 together make up a K_4, they receive 4 distinct colors. Without any loss of generality, we can assign them $(1, 2, 3, 4)$ respectively (as they make up a K_4 and we are considering proper coloring). It follows that at least one of the vertices y_1, y_2, y_3 is colored with 5, as otherwise x_0 would not have a unique color in its neighborhood. This means that at least two of the vertices x_1, x_2 and x_3 have 4 distinct colors in their open neighborhoods.

Now, note that G is made of four copies of H, with five vertices a_1, a_2, a_3, a_4, a_5, not corresponding to x_0 (they are not in the middle of H), placed in such a way that every vertex belongs to two copies of H. Altogether, in G there are seven $a_1, a_2, a_3, a_4, a_5, a_6, a_7$ vertices corresponding to one of x_1, x_2, x_3. Since in each copy of H, at least two of them see 4 distinct colors in their open neighborhoods, at least one of the vertices sees all 4 colors at least twice. The fifth color is assigned to the vertex itself and since we are considering proper coloring it cannot be its unique color, therefore we are getting our contradiction. □

7.3 Upper bounds for planar graphs

The first proof showing $\chi_{pCFC_o} \leq 8$, which is the current best bound, was done in [3]. The authors proved it by mainly relying on a concept known as facial closure, facial-conflict-free coloring, vertex partition and 4-color Theorem 1.2.1. In this section however, we will show an original proof of this thesis, proving the same result without relying on the facial closure and facial conflict-free coloring.

Definition 7.3.1. (BFS layering): Let G be a connected graph and $r \in V(G)$ a fixed vertex. The BFS layering of G determined by r is the vertex partition $L = (L_0 = r, L_1, L_2, ...)$ of G such that L_i contains all vertices of G at distance exactly i from r.

Observation 7.3.1. *When considering BFS layering, it holds that for every vertex $v \in L_i$ there are edges only between vertices of the same layer L_i or layers L_{i+1} or L_{i-1}.*

This is because if there was an edge between vertex v from layer L_i to layer L_{i-2}, then vertex v is not at distance i from r, but $i - 1$ and therefore would not belong to the layer L_i.

Observation 7.3.2. *When considering BFS layering, it holds that for every vertex $v \in L_i$, where $i > 0$, there is at least one edge connecting it to the previous layer L_{i-1}.*

Theorem 7.3.1 (Equivalent to [3])**.** *For every planar graph G we have $\chi_{pCFC_o}(G) \leq 8$.*

Proof. Let G be a connected planar graph. First, we select an arbitrary vertex r of G. Next, we consider the BFS-layering L from r. Next, we will construct two planar graphs G_1 and G_2, which we will color using 4-color Theorem 1.2.1.

To construct G_1 we take every even layer in L (we consider layer 0 of r to be even) and remove all edges between the vertices of the same even layer. Next, we contract r into its arbitrary neighbor in L_1. Then we contract every vertex from every even layer into its arbitrary neighbor from the previous layer (we know that such neighbor exists for every vertex thanks to Observation 7.3.2). This gives us a planar graph G_1 which consists of only the odd layers from L.

To construct G_2 we take every odd layer in L and remove all edges between the vertices of the same odd layer. Next, we contract every vertex from every odd layer into its arbitrary neighbor from the previous layer (we know that such a neighbor exists for every vertex thanks to Observation 7.3.2). This gives us a planar graph G_2 which consists of only the even layers from L.

Finally, we properly color G_1 using colors $\{1, 2, 3, 4\}$ and G_2 using colors $\{5, 6, 7, 8\}$. We argue that this coloring is a valid $pCFC_o$.

First, we observe that this coloring is proper. Since vertices of the same layer were colored in G_x using 4-color Theorem 1.2.1 and since its adjacent layers have different 4 colors and we use the Observation 7.3.1 we can see that there indeed are no two vertices of the same color next to each other.

To show that this is a valid CFC_o we can argue that every vertex has its unique color on the vertex it was contracted into (by contracting a vertex v into its n neighbor u we make all neighbors of v adjacent to u. Then by using proper coloring, and extending it to the original graph the color assigned to u is unique in the neighborhood of v). Since we know that every vertex was contracted into a different layer than the one it belonged to, we also know that its unique color is from a different set of colors than the set of colors of its layer. □

8 Conclusion

Our thesis has started by introducing the basic concepts of graph theory, followed by definitions of various variants of conflict-free coloring. In Chapter 2 we have also briefly mentioned studies done on CFC and subjects related to it. Mainly we have discussed and looked into studies about bounds on chromatic numbers for these colorings and showed a comprehensive list of the best bounds. It was pointed out that many of the bounds are yet to be tight, so there is a lot of potential future research to be done and many open problems.

In Chapter 4 we have managed to solve one of the questions by proving the tightness of the bound for CFC_c on planar graphs. More specifically we have improved the lower bound from 3 to 4 by giving an example of a planar graph that does not have a valid CFC_c using 3 colors.

In [3] however, it was conjectured that most of the lower bounds are already the best that can be, so it was recommended to be more focused on improving the upper bounds instead.

We have also shown proof originating from [6] that may still be improvable or valuable in proving other bounds. We have shown a few ideas about how the construction in this proof could possibly lead to some new results.

The biggest room for improvement may be in bounds concerning $pCFO_o$ as this variant of coloring was only recently introduced in [3]. Despite the fact that it is a relatively new concept, quite a bit of research has been done on it already (see [16, 17]), but none of them have managed to definitively improve the bounds yet.

In Chapters 6 and 7 we have briefly studied these bounds for $pCFO_c$ as well as $pCFO_o$, even showing an original proof for one of the bounds more precisely we proved $\chi_{pCFC_o} \leq 8$. This bound is however still far from being optimal as the lower bound is 6.

In spite of there being a lot of open questions, the research into these bounds is limited and becoming harder as they approach tightness.

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